

Assignment 2 Team 10 Ops Brief

Team Members: Ethan Quinn, Jack Strabala, Tomas Turner

Regression: how MAE translates to staffing/gate planning

For our project, we decided to use the flights dataset, where we would train and evaluate models on information about diverted flights. From a real-world perspective, the MAE translates to how far off your predicted delay is from the actual delay on average. So let's say that our MAE is 10 min, that means that our average delay time is off by 10 min. If we were to think that the diversion duration was 20 min, it would actually be 30. In operations, and more importantly in air traffic operations, 10 minutes can be very costly, it can affect whether the staff is reading to land the plane, whether or not the gate is free, or how buffers are scheduled. In a staffing sense, the smaller the MAE the more reliable schedules you have, and the fewer idle ground crews or sudden overtime needs. For gate planning, it can get a bit more complicated. Due to the nature of airports, there are central controls and plans as to where every plane will land and take off, and not having a correct MAE can cause a lot of chaos throughout the airport. Missed communication on timing can delay flights, keep them from taking off, or make dispatchers have to reassign gates to the airplanes coming in.

Classification: the risk matrix (FP vs FN) and why your threshold is appropriate

We trained a logistic regression classifier to predict whether or not a flight will be diverted or not. At our default threshold of 0.5, our confusion matrix basically does not predict any diversions. This is because diversions are extremely rare so the model predicting not diverted each time, is correct a majority of the time. However, in this scenario False Positives likely lead to some soft planning that likely has low direct cost. On the other hand, False Negatives means that there was no preparations and the

diversion still happens so staff must act reactively. This could lead to potential routing issues and gate chaos. Therefore, in this scenario, FN is more costly than FP for predicting diversions. To better try to decrease FNs, we reduced the threshold to 0.25, to identify more flights being diverted. While this will increase some FPs, the tradeoff of decreasing the number of FNs is worth it.

Feature engineering wins and future ideas.

The engineered model, unlike the baseline model, allows us to use the TRANSFORM clause to create new and more informative features from the raw data. For our model more specifically, we created the route dataset by concatenating the origin and the destination, and bucketize departure delay into either “early”, “on-time”, ‘minor’, ‘moderate’, or ‘major’. The baseline model on the other hand, only uses raw input features directly, such as distance, carrier, dep_delay, and others. Although we did engineer our model, we got similar or even worse results compared to the baseline. In both models, our recall and precision scores were almost 0.00, our accuracy was very slightly higher, and interestingly, the roc_auc and log_loss both got worse with the engineered model. In actuality there were little wins for the feature engineering, but we came up with some ideas that we could implement in the future to get more wins. The first idea that we had was to figure out more advanced features, like creating interaction terms, or polynomial features for the numerical variables. We also thought of some temporal feature indicators like holiday or peak travel seasons, we believe that there might be some correlation between that and the amount of diversions. We could also do some hyperparameter tuning for the model, or increase the data volume for training. Currently, our limit was at 10,000 because of certain cost parameters that Colab did not allow us to override, so maybe also having the model be able to train on more data would make it perform better.